Southwest Advanced Prototyping Hub
Arizona State University

intel

Official Problem Brief

Semiconductor Solutions Challenge 2026

Problem A:
Small Sample Learning for Defect Classification

Developed in partnership with Intel Corporation

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This document outlines the technical problem you will address, including the problem overview, objectives, scope, deliverables, and resources.

Each problem represents a real-world challenge currently faced by leaders in semiconductor manufacturing, electronic design automation (EDA), and applied AI. Your task is to develop an innovative, technically grounded solution within the competition timeline.

Depending on the specific challenge, students may be expected to propose a well-reasoned conceptual solution, develop a simulation framework, or build and evaluate a working prototype or model. **Please review the deliverables section of your specific problem carefully.**

This competition is designed to simulate real industry problem-solving. Strong solutions will demonstrate:

- Technical rigor
- Clear assumptions and reasoning
- Measurable impact
- Practical feasibility
- Awareness of constraints and trade-offs

Competition Timeline

Final Submission

Participants are required to submit a presentation detailing their solution. The final submission form must be completed by **March 20, 2026 at 5:00 PM MST**. Late submissions will not be reviewed for advancement to the finalist presentation round.

Questions and Support

Contact **swaphubworkforce@asu.edu**.

Problem Overview

Defect classification is essential to semiconductor manufacturing, where accurate and timely detection directly impacts yield, product quality, and time-to-market for advanced technologies used worldwide. In real production environments, labeled defect data is often limited, creating a gap between practical needs and traditional machine learning approaches that rely on large datasets. In this challenge, you will build a model capable of detecting and classifying defects from limited data. Your approach should reflect how a human expert rapidly learns and adapts from just a few examples. Your work has the potential to accelerate yield learning and improve manufacturing efficiency in real-world semiconductor production.

Objectives

1. Enable solution scalability via automation
2. Improve insights into defect categories
3. Improve yield and product quality

Scope

1. Process a gray-scale image in approximately 1 second using the hardware recommended as part of the solution.
2. Handle image sizes up to approximately 1500 × 2500 pixels.
3. Accurately classify images into 8 defect classes, accounting for highly imbalanced data.

Deliverables

1. Develop an application capable of detecting and classifying defects from gray-scale images.
2. Provide evaluation plots showing detection and classification accuracy versus defect class occurrence, using randomized samples.
3. Achieve a target of approximately 85% overall classification accuracy.
4. Demonstrate how quickly the model learns, evaluated on a randomized sequence of held-out images provided by Intel.
5. Include documentation summarizing the model approach, assumptions, handling of class imbalance, and hardware used.

Data, Tools, and Resources

Students may use public datasets, standard machine learning, and AI libraries.

Intel has provided sample images for students to work with. The sample images are located in the dataset folder and dataset.zip file. Since both contain identical data, **please use only one**.